Decidability 04

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The Set of Real Numbers is Uncountable

- A **real number** $\mathbb{R}$ is a number with decimal representation
  - $\pi = 3.14159265\ldots$
  - $345 = 345.000\ldots$
  - $1.23 = 1.23000\ldots$
- To show that a set $\mathbb{R}$ is not countable, we generally use proof by contradiction
  - Assume that $\mathbb{R}$ is countable
  - $\mathbb{R}$ is countable, there is a correspondence between $\mathbb{N}$ and $\mathbb{R}$
  - Since there is a correspondence between $\mathbb{N}$ and $\mathbb{R}$, there are infinitely many correspondence between $\mathbb{N}$ and $\mathbb{R}$
  - This allows us to fill the right column of the correspondence table any way we want (no duplicate):

$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$	$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$	$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$
1	?	1	$1.000\ldots$	1	$1.414\ldots$
2	?	2	$2.000\ldots$	2	$3.141\ldots$
3	?	3	$3.000\ldots$	3	$2.222\ldots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

  - Always results in a valid correspondence

Diagonalization

- Consider an assignment of the function $f : \mathbb{N} \to \mathbb{R}$ as shown below:

$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$
1	$3.141592\ldots$
2	$55.555555\ldots$
3	$0.123456\ldots$
4	$0.500000\ldots$
5	$1.414213\ldots$
$\vdots$	$\vdots$

- This is one of infinitely many valid correspondences
- We need to find an element $x \in \mathbb{R}$ where $f(n) \neq x$ for any $n \in \mathbb{N}$
  - If we can find such x, the above table is not a valid correspondence (not onto)

Diagonalization

- Consider an assignment of the function $f : \mathbb{N} \to \mathbb{R}$ as shown below:

$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$
1	$3.\boxed{1}41592\ldots$
2	$55.5\boxed{5}5555\ldots$
3	$0.12\boxed{3}456\ldots$
4	$0.500\boxed{0}00\ldots$
5	$1.4142\boxed{1}3\ldots$
$\vdots$	$\vdots$

- From the above table:
  - We choose the 1st digit after the decimal point of the real number that corresponds with 1
  - We choose the 2nd digit after the decimal point of the real number that corresponds with 2
  - We choose the 3rd digit after the decimal point of the real number that corresponds with 3
  - And so on

Diagonalization

$n \in \mathbb{N}$	$f(n) \in \mathbb{R}$
1	$3.\boxed{1}41592\ldots$
2	$55.5\boxed{5}5555\ldots$
3	$0.12\boxed{3}456\ldots$
4	$0.500\boxed{0}00\ldots$
5	$1.4142\boxed{1}3\ldots$
$\vdots$	$\vdots$

$$x = 0.26412\ldots \in \mathbb{R}$$

- Constructing $x \in \mathbb{R}$ starting from the first digit after the decimal point:
  - The 1st digit after the decimal point of $f(1)$ is 1 $\rightsquigarrow$ pick 2
  - The 2nd digit after the decimal point of $f(2)$ is 5 $\rightsquigarrow$ pick 6
  - The 3rd digit after the decimal point of $f(3)$ is 3 $\rightsquigarrow$ pick 4
  - The 4th digit after the decimal point of $f(4)$ is 0 $\rightsquigarrow$ pick 1
  - The 5th digit after the decimal point of $f(4)$ is 0 $\rightsquigarrow$ pick 2
  - And so on
- Obviously, x is a real number ($x \in \mathbb{R}$)

Diagonalization

- Let's recap how we constructed x:
  - If the nth digit after the decimal point of $f(n)$ is y, pick a number that is not equal to y
  - We use that number (not equal to y) as the nth digit after the decimal point of x
- Claim: No $n \in \mathbb{N}$ such that $f(n) = x$
  - Because the nth digit after the decimal point of x will be different that the nth digit after the decimal point of $f(n)$
- The correspondence is not onto
  - Not a valid correspondence
  - $\mathbb{R}$ is uncountable
- This method of constructing x is called **diagonalization**

Σ^* is Countable

- Let $\Sigma = \{0, 1\}$
- How to show that Σ^* is countable
  - We need to show a correspondence between $\mathbb{N}$ and Σ^*
  - Let's list element of Σ^* from the shortest one

$n \in \mathbb{N}$	$f(n) \in \Sigma^*$
1	ε
2	0
3	1
4	00
5	01
6	10
7	11
8	000
$\vdots$	$\vdots$

- By listing from the shortest, all strings will be on the right column
- Thus, one-to-one and onto (a valid correspondence)
- Σ^* is countable

Σ^* is Countable

- Another way is to explicitly define the function $f : \mathbb{N} \to \Sigma^*$
  - Not necessary
  - The table from the previous slide is good enough
- Example, let $\Sigma = \{0, 1\}$
  - $f(1) = \varepsilon$
  - $f(n) =$ binary representation of $n - 2^{\lfloor \log_2(n) \rfloor}$ with length $\lfloor \log_2(n) \rfloor$

$\mathcal{N}$	$\log_2 n$	$\lfloor \log_2 n \rfloor$	$2^{\lfloor \log_2(n) \rfloor}$	$n - 2^{\lfloor \log_2(n) \rfloor}$	Σ^*
1					ε
2	1	1	2	0	0
3	1.585	1	2	1	1
4	2	2	4	0	00
5	2.322	2	4	1	01
6	2.585	2	4	2	10
7	2.807	2	4	3	11
8	3	3	8	0	000
9	3.169	3	8	1	001

- Σ^* is countable

Infinite Binary Sequence is Uncountable

- Let $\mathcal{B}$ be a set of **infinite binary sequence**:

$$\mathcal{B} = \{x_1 x_2 \dots x_\infty \mid x_i = 0 \text{ or } x_i = 1\}.$$

- Show that $\mathcal{B}$ is uncountable
- Again, we need to use proof by contradiction and diagonalization:
  - Assume that $\mathcal{B}$ is countable
  - Thus, there exists a correspondence $f : \mathbb{N} \to \mathcal{B}$
  - Since there is one, there are infinitely many correspondences
  - This is one of infinite many valid correspondences (based on the assumption that $\mathcal{B}$ is countable):

$n \in \mathbb{N}$	$f(n) \in \mathcal{B}$
1	$0100101\dots$
2	$1101011\dots$
3	$1010101\dots$
4	$0010011\dots$
5	$0001000\dots$
$\vdots$	$\vdots$

Diagonalization Example for Infinite Binary Sequence

$n \in \mathbb{N}$	$f(n) \in \mathcal{B}$
1	$\boxed{0}100101\ldots$
2	$1\boxed{1}01011\ldots$
3	$10\boxed{1}0101\ldots$
4	$001\boxed{0}011\ldots$
5	$0001\boxed{0}00\ldots$
$\vdots$	$\vdots$

- From the above correspondence:
  - Pick the 1st bit (from left) of $f(1)$
  - Pick the 2nd bit (from left) of $f(2)$
  - Pick the 3rd bit (from left) of $f(3)$
  - Pick the 4th bit (from left) of $f(4)$
  - $\vdots$
  - Pick the nth bit (from left) of $f(n)$
  - $\vdots$

Diagonalization Example for Infinite Binary Sequence

$n \in \mathbb{N}$	$f(n) \in \mathcal{B}$
1	$\boxed{0}100101\ldots$
2	$1\boxed{1}01011\ldots$
3	$10\boxed{1}0101\ldots$
4	$001\boxed{0}011\ldots$
5	$0001\boxed{0}00\ldots$
$\vdots$	$\vdots$

$$x = 10011\ldots \in \mathcal{B}$$

- Construct $x \in \mathcal{B}$ starting from the first bit (from left)
  - The 1st bit (from left) of $f(1)$ is 0 $\rightsquigarrow$ pick 1
  - The 2nd bit (from left) of $f(2)$ is 1 $\rightsquigarrow$ pick 0
  - The 3rd bit (from left) of $f(3)$ is 1 $\rightsquigarrow$ pick 0
  - The 4th bit (from left) of $f(4)$ is 0 $\rightsquigarrow$ pick 1
  - The 5th bit (from left) of $f(4)$ is 0 $\rightsquigarrow$ pick 1
  - $\vdots$
  - The nth bit (from left) of $f(n)$ is y $\rightsquigarrow$ pick $\neg y$
  - $\vdots$
- Obviously, x is an infinite binary sequence

Diagonalization Example for Infinite Binary Sequence

- When we construct x:
  - If the nth bit (from left) of $f(n)$ is y, pick a number that is not equal to y
    - If it is 0, pick 1
    - If it is 1, pick 0
  - We use that number as the nth bit (from left) of x
- Claim: No $n \in \mathbb{N}$ such that $f(n) = x$
  - Because the nth digit bit (from left) of x will be different that the nth bit (from left) of $f(n)$
- The correspondence is not onto
  - Not a valid correspondence
  - The set of all infinite binary sequences ($\mathcal{B}$) is uncountable

The Set of Turing Machines in Countable

- Each Turing machine has its own encoding $\langle M \rangle$
  - A string representation of TM M
  - A string is a finite sequence of symbols over a Σ
- $\langle M \rangle$ for every machine has a finite length
- Similar to Σ^*, we can list all Turing machines in their encoding form starting from the shortest length
- Therefore, the set of Turing Machines is countable

Set of All Languages is Uncountable

- Diagonalization works well when all elements are infinite
  - We need to pick the nth of something
    - The nth digit after the decimal point
    - The nth bit (from left)
  - If all elements are infinite long, it is guaranteed to have the nth of something
- But some languages are finite and some are infinite
  - Diagonalization does not work well for proving that the set of all languages is uncountable

Same Size, Countable, and Uncountable

- The following are what we have so far:
  - The set of all Turing machine is countable
  - The set of all infinite binary sequences is uncountable
- We know that an uncountable set is larger than a countable set
- We need to show that the set of all languages has the same size as the set of all infinite binary sequences
- To show that two sets have the same size, we need to show a correspondence
- We need a correspondence between the set of all languages and the set of all infinite binary sequences

The Set of All Languages is Uncountable

- We can list the set of all strings over $\Sigma = \{0, 1\}$ starting from the shortest string as follows:

$$\{\varepsilon, 0, 1, 00, 01, 10, 11, 000, 001, \dots\}$$

- Given a language (either finite or infinite), we can also list the language starting from the shortest string
  - The set of all strings that start with a 1

$$\{1, 10, 11, 100, 101, 110, 111, 1000, \dots\}$$

  - The set of all strings of length 2

$$\{00, 01, 10, 11\}$$

- Note that we use dictionary order for those with the same length

The Set of All Languages is Uncountable

- Given a language, we can always list their strings to align with strings in Σ^*
  - The set of all strings of length 2

$$\begin{array}{rl} \Sigma^* & = \{\ \varepsilon,\ 0,\ 1,\ 00,\ 01,\ 10,\ 11,\ 000,\ 001,\ \ldots\} \\ A & = \{\ \quad\quad\quad\ 00,\ 01,\ 10,\ 11\quad\quad\quad\ \} \end{array}$$

  - The set of all strings that start with a 1

$$\begin{array}{rl} \Sigma^* & = \{\ \varepsilon,\ 0,\ 1,\ 00,\ 01,\ 10,\ 11,\ 000,\ 001,\ \ldots\} \\ A & = \{\ \quad\quad\ 1,\ \quad\quad\ 10,\ 11,\ \quad\quad\ \ldots\} \end{array}$$

  - The set of all strings that start with a 0

$$\begin{array}{rl} \Sigma^* & = \{\ \varepsilon,\ 0,\ 1,\ 00,\ 01,\ 10,\ 11,\ 000,\ 001,\ \ldots\} \\ A & = \{\ \quad\ 0,\ \quad\ 00,\ 01,\ \quad\quad\ 000,\ 001,\ \ldots\} \end{array}$$

The Set of All Languages is Uncountable

- When we align them as in previous slide, we can convert the alignment into an infinite binary sequence $\mathcal{X}_A$:

$$\begin{array}{rcccccccccccc} \Sigma^* & = \{ & \varepsilon, & 0, & 1, & 00, & 01, & 10, & 11, & 000, & 001, & \ldots \} \\ A & = \{ & & 0, & & 00, & 01, & & & 000, & 001, & \ldots \} \\ \mathcal{X}_A & = & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & \ldots \end{array}$$

  - $\mathcal{X}_A$ is the infinite binary sequence that corresponds with the language A
  - $\mathcal{X}_A$ is a member of the set of all infinite binary sequences $\mathcal{B}$
- The above shows that for every language A, it corresponds to $\mathcal{X}_A \in \mathcal{B}$
  - The set of all languages has the same size as the set of all infinite binary sequences
  - Since the set of all infinite binary sequences is uncountable, the set of all languages is uncountable

Some Languages are not Turing-recognzable

- Recall that
  - the set of all Turing machines is **countable**
  - the set of all languages is **uncountable**
- The set of all languages is larger than the set of all Turing machine
- Note that one Turing machine only recognizes one language

Corollary 4.18

Some languages are not Turing-recognizable.